Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

A discrete source X emits symbols from a finite alphabet of $M = 6$ symbols

- Compute the maximum entropy that the source can achieve. What is the condition that need to be verified for this to hold?
- By observing a large number of symbols (i.e. $N \to \infty$), the estimated entropy is $\hat{H}(X) = 1.6$ bps. Compute the lower and upper bounds on the average code length of Shannon's coding.
- The estimated entropy rate is $\hat{H}_\infty(X) = 0.9$ bps. Based on this observation, list lossless coding methods that can be used to compress such a source. Select one of them and discuss it in detail.

2 Question - 5 pts

Illustrate the properties of speech signals. Describe the basic ideas and building blocks of the source-system model for speech synthesis. In particular, explain the difference between the short-term and the long-term predictor.

3 Question - 12 pts [Matlab]

Consider a speech segment (load the sample signal `matlab`).

- Compute the prediction coefficients for each 20ms time window (use $p = 10$ coefficients) [Hint: you can use `lpc` or `levinson`]
- Design a suitable quantizer for the prediction coefficients allocating $R = 80$ bits per window [Hint: you can use `ac2rc` and `rc2ac`].
- Synthesize speech using the quantized prediction coefficients and the ideal excitation signal obtained at the encoder
- Plot the original and synthesized speech and compute the SNR

4 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a memoryless source with an infinite alphabet, e.g. $x \in \mathbb{R}$, and $D(R)$ the corresponding rate-distortion curve
   - [T] [F] $D(0) = \infty$
   - [T] [F] $D(R) \geq D_S(R)$, where $D_S(R)$ is Shannon's bound
   - [T] [F] $R(0) = H(X)$
   - [T] [F] $D(R_i) \leq D(R_j)$, if $R_i < R_j$
2. Consider an image coding system
   - [T] [F] The choice of the block size affects coding efficiency
   - [T] [F] Zonal quantization addresses the non-stationary behaviour of natural images
   - [T] [F] The luminance component is downsampled by a factor of 2 in both directions
3. Consider a video coding system and a GOP structure $IPP...I$
   - [T] [F] The coding efficiency improves by increasing the GOP size
   - [T] [F] The computational complexity increases by increasing the GOP size
   - [T] [F] The adoption of P-frames introduces drift